Program 3(b): Parking Management System

```
interface ParkingCharge {
    double calculateCharge();
    void displayParkingDetails();
}

class VehicleParking {
    String vehicleNumber;
    String vehicleType;
    int parkingHours;

    public VehicleParking(String vehicleNumber, String vehicleType, int
parkingHours) {
        this.vehicleNumber = vehicleNumber;
        this.vehicleType = vehicleType;
        this.parkingHours = parkingHours;
    }
}

class TwoWheeler extends VehicleParking implements ParkingCharge {

    public TwoWheeler(String vehicleNumber, int parkingHours) {
        super(vehicleNumber, "Two Wheeler", parkingHours);
    }

   public double calculateCharge() {
        if (parkingHours <= 2) {
            return parkingHours * 20.0;
        } else {
            return (2 * 20.0) + ((parkingHours - 2) * 10.0);
        }
    }

   public void displayParkingDetails() {
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours + " hrs");
        System.out.println("Total Parking Charge: Rs. " + calculateCharge());
        System.out.println("-----------------------------------");
    }
}

class FourWheeler extends VehicleParking implements ParkingCharge {

    public FourWheeler(String vehicleNumber, int parkingHours) {
        super(vehicleNumber, "Four Wheeler", parkingHours);
    }

   public double calculateCharge() {
        if (parkingHours <= 2) {
            return parkingHours * 50.0;
        } else {
            return (2 * 50.0) + ((parkingHours - 2) * 30.0);
        }
    }

    public void displayParkingDetails() {
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours + " hrs");
```

```
        System.out.println("Total Parking Charge: Rs. " + calculateCharge());
        System.out.println("-----------------------------------");
    }
}

public class ParkingManagementSystem {
    public static void main(String[] args) {

        ParkingCharge parkingRef;

        parkingRef = new TwoWheeler("KA-01-AB-1234", 5);
        parkingRef.displayParkingDetails();

        parkingRef = new FourWheeler("KA-05-CD-5678", 4);
        parkingRef.displayParkingDetails();


        System.out.println("Name - Kunal Suryawanshi \nUSN - 1BF25CS132");
    }
}
```

O/P

```
 ----jGRASP exec: java ParkingManagementSystem
Vehicle Number: KA-01-AB-1234
Vehicle Type: Two Wheeler
Parking Hours: 5 hrs
Total Parking Charge: Rs. 70.0
-----------------------------------
Vehicle Number: KA-05-CD-5678
Vehicle Type: Four Wheeler
Parking Hours: 4 hrs
Total Parking Charge: Rs. 160.0
-----------------------------------
Name - Kunal Suryawanshi
USN - 1BF25CS132

 ----jGRASP: Operation complete.
```